Practical 8: Magmatism at destructive plate margins

On successful completion of this practical, you should be able to:

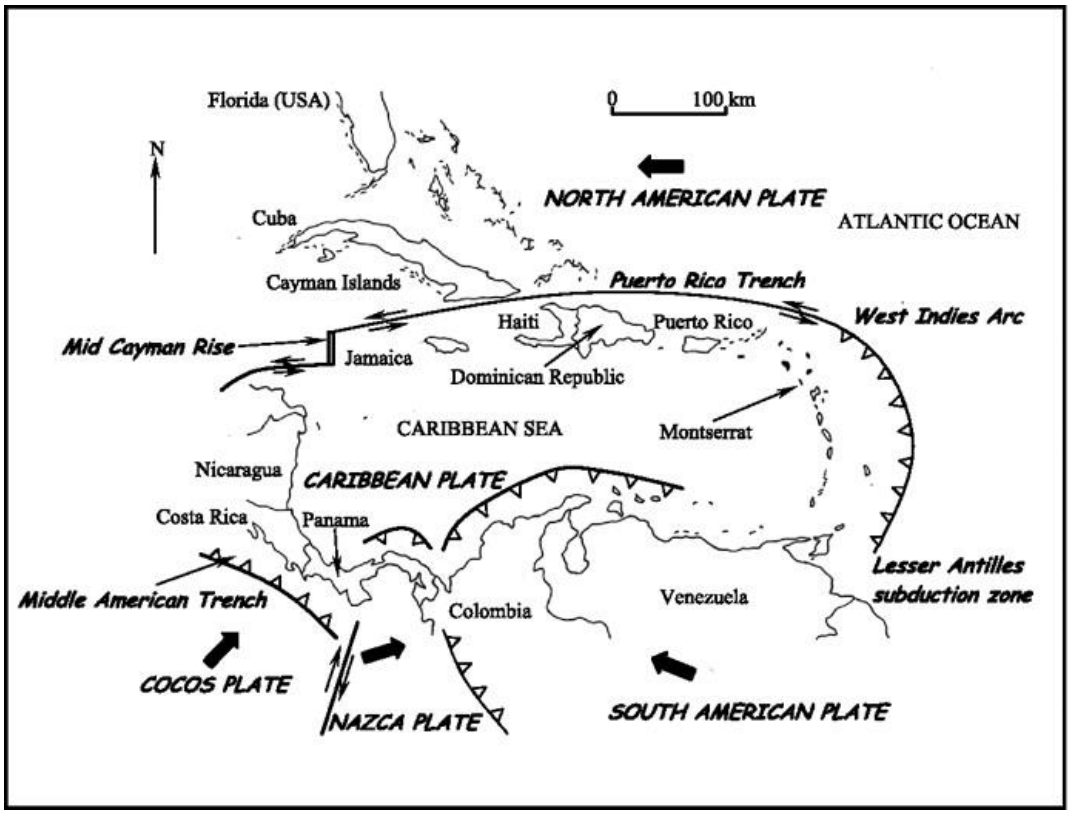
- Recognise and describe the mineralogy and textures of subduction-related igneous rocks
- Explain the concept of disequilibrium
- Identify disequilibrium mineral textures in a thin section
- Interpret magma storage and crystallization conditions using a pressure-temperature phase diagram

You will also practice the following skills:

- Interpret the order of crystallization in complex igneous rocks from observed igneous textures
- Make an accurate, labelled sketch of the mineralogy and texture of a thin section

Part 1 (40 minutes): Andesites from Soufrière Hills Volcano, Montserrat

The island of Montserrat in the West Indies is part of the Caribbean Arc. The island was formed, and continues to grow, due to island arc magmatism caused by subduction of the North American Plate beneath the Caribbean Plate.



Soufrière Hills Volcano (SVH) is named after its sulfur-rich hot springs, or *soufrières*. The predominant magma composition erupted at SVH is andesite. A typical eruption style is for viscous andesitic magma to extrude slowly, forming a lava dome, which then undergoes periodic collapse, generating pyroclastic density currents.

<u>SHV1 or SHV2</u> *Soufrière Hills Volcano, Montserrat*

Sample SHV1 is a representative sample from SHV's lava dome. It contains inclusions (or *enclaves*) of a more mafic magma that mixed with the host magma before eruption. The magma mixing may even have triggered the eruption. You should focus on the host magma.

Sample SHV2 is a breadcrust bomb from a mildly explosive eruption. The outer glassy "crust" forms due to rapid chilling when the magma erupts and comes into contact with air. The hot viscous interior of the breadcrust bomb continues to vesiculate. This causes the bomb to expand like a loaf of bread, creating the distinctive cracked surface.

Table 1. Major element compositions of representative samples from Soufrière Hills Volcano and Hakone Volcano obtained by X-ray fluorescence (XRF) analysis.

Oxide, wt.%	SVH1	H-21†	H-23†
SiO_2	61.28	56.62	67.37
TiO_2	0.58	0.96	0.72
Al_2O_3	17.30	15.49	15.28
Fe_2O_3	6.34	12.46	5.42
MnO	0.19	0.22	0.15
MgO	1.86	3.00	1.20
CaO	7.28	7.33	4.46
Na_2O	3.47	2.95	4.67
K_2O	0.84	0.55	0.92
P_2O_5	0.21	0.08	0.18

†Kuno H (1960) Petrology of Hakone volcano and the adjacent areas, Japan. Geol Soc Am Bull 61:957-1020.

The chemical composition of the host magma in SVH1 is given in Table 1. Use the TAS diagram to classify this rock. (SVH2 has a very similar chemical composition.)

SVH1 is an andesite.

SVH1 and SVH2 both contain microphenocrysts of opaque magnetite. What are the three phenocryst minerals?

The phenocrysts are hornblende, plagioclase, orthopyroxene. Some thin sections contain microphenocrysts of augite.

Describe and sketch the disequilibrium textures shown by:

a) the colourless, low-relief phenocrysts (two textures)

Plagioclase: Most crystals have oscillatory zoning, probably formed in response to changing magma composition and changing dissolved water content in the magma. Some crystals have sieve-textured zones with glass inclusions.

b) the strongly coloured phenocrysts (two textures)

Hornblende: Crystals are often rounded. Some have thin rims of fine-grained opaque oxides (opacite). Some hornblendes have a much thicker reaction rim that is an intergrowth of cpx + opx + plg + mgt.

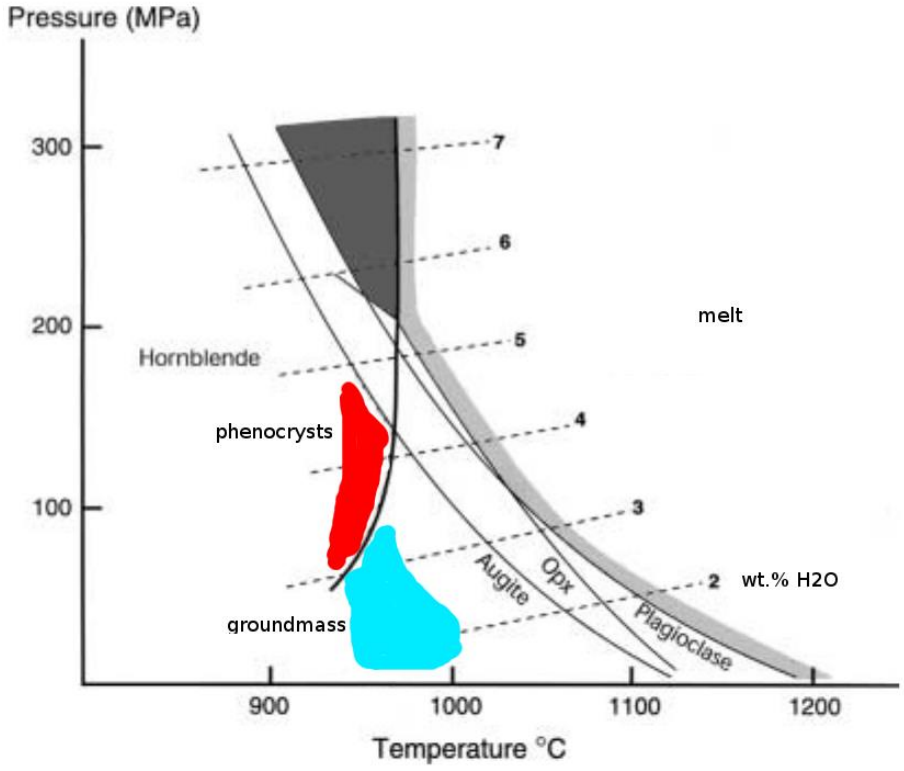
c) any other disequilibrium textures you notice (variable between thin sections – don't worry if you don't find any).

Most of the other phenocrysts don't show any obvious disequilibrium features. You may have noticed one or both of the following:

- Some thin sections contain small crystals of resorbed quartz. The disequilibrium feature is their rounded shape, which suggests that they have partially dissolved into the magma.
- In some thin sections there are opx crystals surrounded by a thin rim of augite (cpx). The opx is a good nucleation site for the cpx because it has a similar crystallographic structure. This tells us that the cpx crystallized later than the opx. This is not a normal crystallisation sequence: in typical calc-alkaline volcanic rocks, augite is more likely to occur in less evolved compositions, and opx in more evolved compositions. The unusual crystallization sequence suggests that the magma was probably heated just before eruption, causing the cpx rims to crystallise. The heating event may have caused some of the other disequilibrium textures, e.g. the plagioclase sieve textures. The most likely source of this heating is the injection of hot mafic magma – and the mafic inclusions in the thin section provide evidence that a mafic injection occurred just before eruption. (If the mafic injection had occurred a long time before eruption, the andesitic and mafic magmas would have mixed and homogenised, and the mixed magma would have cooled down a little.)

If you wish, you can read more in this paper:
Devine JD *et al*. (1998) Petrologic evidence for pre-eruptive pressure-temperature conditions, and recent reheating, of andesitic magma erupting at the Soufriere Hills Volcano, Montserrat, WI. Geophys Res Lett 25:3669-3672, https://doi.org/10.1029/98GL01330.

The diagram below shows a simplified phase diagram for andesitic magmas. Indicate the range of potential pressure-temperature conditions under which the phenocryst assemblage and the groundmass assemblage could have crystallized.



The phenocryst assemblage contains hornblende, so it must have crystallized within the hornblende stability field (indicated in red). The groundmass assemblage does not contain hornblende, so it must have crystallized outside the hornblende stability field. The blue field shows the potential range of crystallization pressures and temperatures, but we can predict that the groundmass crystals probably formed at very low pressure, when the magma was at (or very close to) the surface.

Does the phase diagram support the hypothesis that there was an injection of mafic magma prior to the eruption?

Yes, the phase diagram can support the hypothesis of a mafic injection of magma prior to eruption. The mafic magma would have been hotter than the andesite resident in the magma chamber, so a mafic injection would heat the magma just before eruption. The phase diagram does suggest that the groundmass crystals could have formed at higher temperatures than the phenocrysts, so this is consistent with a mafic injection heating the andesite before it erupted.

Use the phase diagram to interpret how the disequilibrium textures in the strongly coloured mineral might have formed. (Ask a demonstrator if you are not sure.)

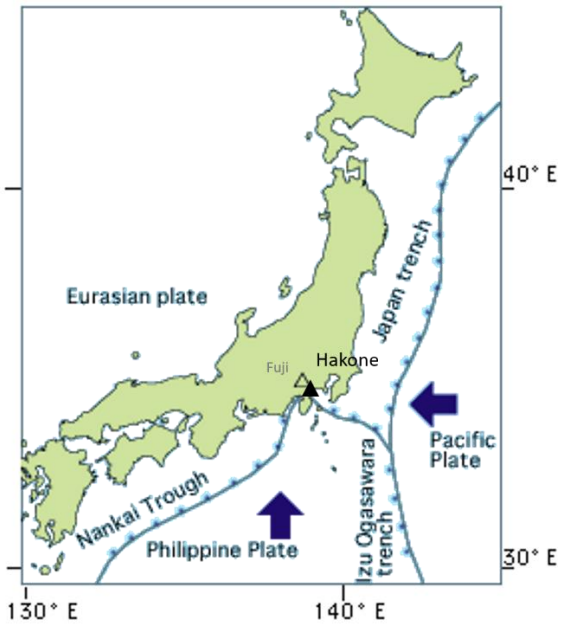
The phase diagram shows that hornblende crystallizes at relatively high pressure, but it is not stable at lower pressures. The thick polymineralic reaction rims of aug + opx + plg could be formed if the hornblende crystals ascended through the magmatic system and were stored in a shallow magma chamber, outside the hornblende stability field, before eruption. This would cause the hornblende to break down and form the reaction rims. The hornblende textures therefore allow us to hypothesize that there were least two magma chambers beneath SVH.

The hornblende crystals with very thin opacite rims have not undergone significant mineral breakdown. These crystals must have ascended rapidly from the deeper magma chamber to the surface, and were never stored in shallower magma chamber. One possibility is that these crystals were carried in the mafic injection. The opacite rims probably formed due to oxidation while the magma was cooling in the lava dome.

Key take-home messages:
(1) Reaction rim thickness is linked to the decompression rate;
(2) The minerals present in the reaction rim can tell us something about the magma storage depth.

Part 2 (40 minutes): Andesites from Hakone, Japan

Hakone volcano is located 90 km southwest of Tokyo in the central part of Honshu Island, Japan. The regional tectonic setting is a complex convergence of four tectonic plates, where the Pacific and Philippine Sea plates are subducting beneath the North American and Eurasian plates.



Hakone is a caldera volcano. The products of ancient basaltic and andesitic stratovolcanoes form the caldera rim. Within the caldera are several younger cones. The oldest cones (130-80 ka) erupted andesite to rhyolite magmas, but the younger central cones (40 ka to present) erupt only andesite.

The chemical compositions of two samples from Hakone volcano are given in Table 1. Use the TAS diagram to classify these two samples from Hakone volcano. Composition H-21 most closely corresponds to the thin sections you will examine.

15391 or 15386 *Hakone, Japan*

This rock contains microphenocrysts of magnetite. What are the other three phenocryst minerals? (One of the phenocrysts is challenging to identify in some of the thin sections; ask a demonstrator if you can't find it.)

The phenocrysts are plagioclase, augite (very pale brown, some simple twinning), orthopyroxene (sometimes rimmed by augite). The opx is challenging to identify and is not obvious in all of the thin sections. It is sometimes surrounded by a rim of augite.

How would you describe the texture of this rock?

The texture is glomeroporphyritic: the phenocrysts often occur in clumps where they have nucleated together.

Describe and sketch the disequilibrium textures you observe in the plagioclase phenocrysts. What process is likely to have produced these textures?

The disequilibrium textures in plagioclase are oscillatory zoning and sieve textures - often both in the same crystal. Oscillatory zoning probably reflects fluctuations in magma water content while the plagioclase was growing. The sieve textures with glass inclusions indicate a period of dissolution at the edge of the crystal followed by rapid growth that traps tiny pockets of magma inside the crystal. Both textures can be produced by episodes of magma recharge by new, hot magma.

Briefly examine the groundmass. You should be able to identify magnetite and laths of plagioclase. The groundmass also contains cpx, opx, glass, and accessory apatite.

Suggest an order of crystallization for this rock, using evidence from the textures you have observed.

Plag is often found as inclusions within pyroxene. Plag is also found around the edges of px crystals, and both plag and px are found in the groundmass. So, plag must have started crystallizing before px, but they both continued to crystallize right up until the time of the eruption. Magnetite is found as inclusions in px, and around the edges of plag; but sometimes plag is found around the edges of mgt – so these three minerals were probably all crystallizing at the same time. Apatite is only found in the groundmass so it must have started crystallizing at a very late stage.

The key point is that the order of crystallization is not a simple linear sequence: multiple minerals are crystallizing at the same time. The textural evidence helps us to estimate when each mineral started and stopped crystallizing.

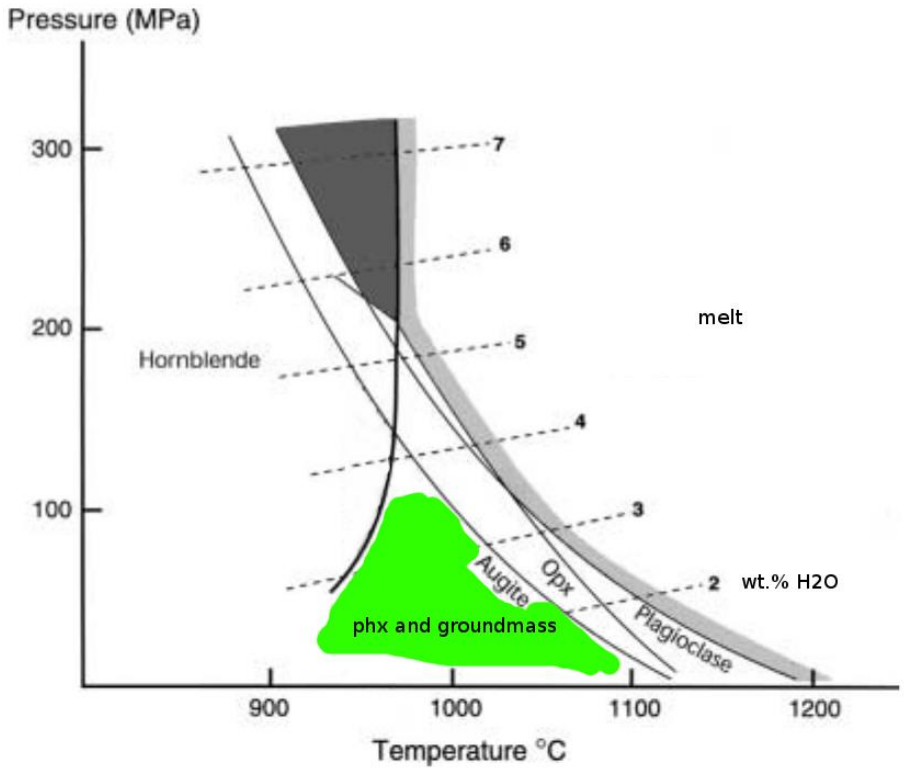
15380 *Hakone, Japan*
This rock contains microphenocrysts of magnetite. The phenocryst minerals are plagioclase, orthopyroxene and augite (not much) – check that you can find and identify these minerals. The groundmass contains plagioclase, clinopyroxene, glass, and accessory apatite.

Examine the abundant brown specks in the plagioclase phenocrysts. These are glass inclusions. They occur either in the interior of the crystal or in a concentric band within the crystal, and they are often elongated parallel to the cleavage. They are probably the result of resorption of phenocrysts which were out of equilibrium with the melt for a period before eruption. The plagioclase rims probably grew as the rock was quenched during eruption.

Do any other phenocrysts show evidence of disequilibrium, and if so, what?

The opx crystals are quite rounded, suggesting that they have partially resorbed into the melt.

Use the phase diagram on page 3 to predict the range of possible crystallization pressures and temperatures for the Hakone andesites. How do these pressures and temperatures compare with the SVH andesites?



The Hakone andesites contain plag + opx + aug as phenocrysts and in the groundmass. They don't contain an hornblende so must plot to the right of the hornblende-in phase boundary. The order of crystallization suggests that plag was first to crystallize, so the magma cannot have been crystallized at pressure greater than ~110 MPa. The possible range is indicated in green.

The phase diagram suggests that the Hakone andesite magma was hotter than the SVH andesite magma, and that it could have been stored at lower pressure than the SVH magmas.

Part 3 (30 minutes): Dacites and rhyolites

182 *Dacite, Ullum-Zonda, Argentina (part of the southern section of the Andes continental arc)*
As well as magnetite, there are four phenocryst minerals in this rock. What are they? Is the phenocryst assemblage consistent with the classification of dacite?

Hornblende, plagioclase, biotite, quartz. Plagioclase and >20% quartz are essential minerals for dacite. Alkali feldspar is missing; however, it may be present in the groundmass. Hornblende and biotite are type minerals for dacites.

What evidence of disequilibrium can you identify in the phenocrysts?

The quartz is rounded. Plagioclase shows resorption and oscillatory zoning. Hornblende shows compositional zoning. Biotite often has rounded ends. All these textures are evidence of disequilibrium.

Use your petrographic observations to suggest a name for this rock.

A suitable rock name would be hornblende dacite.

16684 or 125 *Rhyolite, Taupo, New Zealand*
The groundmass texture in this sample is very characteristic of the deposits from an explosive eruption, and it enables us to say a lot about the eruption style of this rhyolite. We will look at this in more detail in a future practical. Identify the two main felsic phenocrysts, and the rarer mafic minerals. Is the phenocryst assemblage consistent with a rhyolitic magma?

The principal phenocrysts are quartz, plagioclase, magnetite and hornblende. Some thin sections also contain biotite (particularly 16684) and augite.

To be fully consistent with a rhyolite classification the rock should contain sanidine (alkali feldspar). Some of the small simple-twinned crystals are sanidine, but they are not easy to spot because they look very similar to the small plagioclase crystals.

D67 *Quartz diorite, locality unknown*
This is a two-pyroxene quartz diorite (classified using the QAPF classification scheme). This is the plutonic equivalent of an andesite such as 15391.

Bonus digital resources

Here are some Virtual Microscope thin sections of igneous rocks from subduction zones. They contain many of the same minerals and textures as the thin sections you have examined during the practical class.

MontE1a - Andesite - Soufrière Hills Volcano, Montserrat
I16 - Hornblende dacite tuff - Almeria, Spain
I10 - Olivine Andesite - New Zealand
I11 - Pyroxene Andesite - Hungary
Hornblende Porphyrite - Valparaiso - Chile